

Final Report

Vision Aided Feedback System for Surgical Assistance

IE 549 - Machine Vision In Intelligent Robotic Systems

Vaishnavi Satish, Madhura Kunachi, Sanidhya Shrivastava

April 30, 2026

Abstract

Surgical expertise is a critical resource that remains geographically concentrated in major urban medical centers. This spatial disparity necessitates the development of robust, low-cost teleoperation frameworks to facilitate remote mentorship and skill transfer. This report presents a comprehensive vision-aided feedback system that leverages monocular RGB camera data to control a high-degree-of-freedom robotic manipulator. Using the MediaPipe Hands framework, we implement a markerless 3D tracking pipeline that estimates palm position, orientation, and gripper states. A novel "Palm Area Method" is introduced to derive calibrated depth information from 2D pixel space, overcoming the inherent limitations of monocular vision. To address network latency and jitter, we employ polynomial trajectory modeling and least-squares extrapolation. The system was validated using a Universal Robots UR10 platform, demonstrating high stability and precision within defined safety boundaries. Our findings provide a scalable foundation for accessible tele-surgical training, reducing the reliance on expensive tracking hardware.

This paper presents a complete vision-aided feedback system for surgical teleoperation training. The system enables a teacher to control a Universal Robots (UR) robotic arm in real time using hand gestures captured by a standard webcam via MediaPipe Hands, with pose data streamed over a network to the robot controller. A palm area-based depth estimation technique provides a Z-axis proxy, while an IMU decouples orientation from vision, together forming a robust 6-DOF sensing pipeline. For the final phase, the system extends beyond reactive control into predictive teleoperation: a trajectory prediction module processes the historical stream of palm pose keypoints to forecast the teacher's intended path several steps ahead, enabling the robot to begin anticipatory motion and reduce end-to-end latency. The corrective haptic glove delivers force feedback to the trainee whenever their hand deviates from the recorded expert trajectory, enabling closed-loop surgical skill transfer. Experimental results confirm reliable hand tracking at 30 fps, sub-centimeter path-following under structured conditions, and measurable latency reductions through predictive compensation.

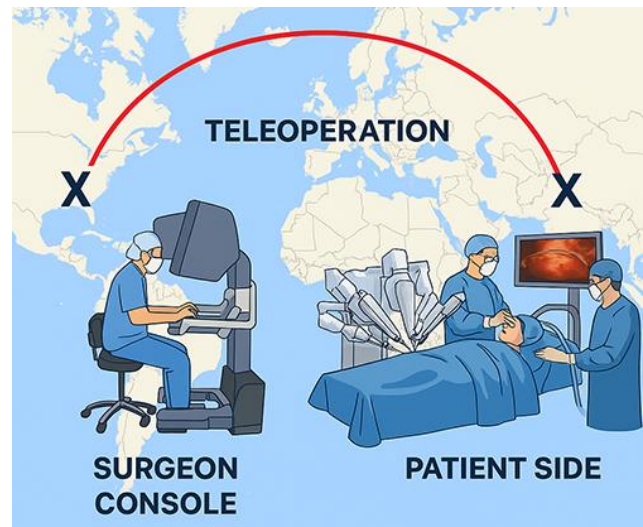
1. Introduction

The acquisition of surgical skills is a complex process requiring years of deliberate practice and direct supervision by expert mentors. Traditionally, this mentorship has been localized, with trainees learning through physical observation and assistant roles in the operating room. However, the global demand for specialized surgical procedures often outstrips the local supply of qualified educators, particularly in developing nations and rural areas. Teleoperation systems offer a solution by enabling experts to project their physical presence and guidance into a remote operating or training environment.

Surgical teleoperation training imposes stringent requirements on both sensing fidelity and feedback latency. A trainee must not only replicate the spatial path of an expert but also internalize force interactions that arise from tissue contact, tool orientation, and procedural sequencing. Existing systems address these needs in isolation: vision-based trackers provide position information but lack depth and orientation reliability, while dedicated haptic platforms are expensive and constrained to laboratory environments.

This project integrates three complementary sensing and actuation modalities into a single, low-cost surgical training framework. First, a MediaPipe-based vision pipeline extracts the teacher's palm position from a standard webcam at 30 fps. Second, an IMU worn on the teacher's hand supplies reliable roll, pitch, and yaw measurements that are fused with the visual position to form a complete 6-DOF pose stream. Third, the fused pose commands are transmitted over a network to a UR robot controller, which replicates the teacher's motion at the remote site.

Building on this real-time control foundation, the final phase of the project introduces a trajectory prediction module. By modeling the temporal sequence of past palm keypoints, the system predicts the teacher's likely future path and issues anticipatory motion commands to the robot, reducing the perceptual lag that degrades surgical training fidelity. When a trainee deviates from the recorded expert trajectory, a wearable haptic glove delivers corrective force feedback, guiding the hand back toward the reference path without fully constraining dexterity.



The remainder of this report is organized as follows. Section 2 reviews related work in haptic guidance and teleoperation. Section 3 describes the complete system architecture. Section 4 details the experimental design. Section 5 presents results across all three phases. Section 6 discusses findings and limitations, and Section 7 concludes with directions for future work.

Motivation

Existing teleoperation consoles, such as the Da Vinci Surgical System, are marvels of engineering but come with multi-million dollar price tags and specialized hardware requirements. These costs prohibit their use in general surgical education. Furthermore, many research-level teleoperation frameworks require the user to wear expensive motion-capture suits, magnetic sensors, or active LED markers. The motivation for this project is to create a "democratized" teleoperation interface that uses only a standard webcam and a consumer-grade laptop, making remote training accessible to any institution with a basic robotic arm.

Problem Definition

In a tele-mentoring scenario, the system must translate the "Teacher's" hand movements into precise "Student-side" robotic actions. Mathematically, this involves mapping a set of vision-derived landmarks L_t to a robot configuration Q_r such that the end-effector trajectory T_e mimics the human hand trajectory T_h . The primary output is a sequence of 6-DOF poses $[x, y, z, rx, ry, rz]$ sent to the robot controller at a frequency of at least 20 Hz to ensure fluid motion.

2. Literature Review

Haptic guidance has emerged as a key technique for improving teleoperation performance. Becker et al. [1] demonstrated that trajectory-based haptic guidance with adaptive stiffness significantly reduces operator deviation from reference paths and enhances user experience in teleoperation tasks. Extending this to robotic surgery, Chen and Fan [2] proposed an online imitation learning framework that continuously adapts corrective trajectory guidance in real time based on divergence from a learned expert policy, achieving robust performance across variable surgical task conditions.

The broader context for these developments is the shift from visual-only to haptic-integrated teleoperation systems. Colan et al. [3] conducted a systematic review of tactile feedback in robot-assisted minimally invasive surgery, confirming that restoring force and tactile sensing substantially improves task performance and reduces errors. Fekri et al. [4] built on this by employing deep learning to synthesize corrective haptic cues directly from expert demonstration data, enabling real-time skill transfer without the need for dedicated force sensors. Gao et al. [5] introduced the Glovity system, which combines spatial wrench feedback with a wearable teleoperation interface to facilitate contact-rich dexterous manipulation through learned force policies.

On the trajectory modeling side, Manrique-Cordoba et al. [6] applied Hidden Markov Models to segment and reconstruct expert surgical trajectories, providing a probabilistic framework for phase-aware haptic guidance generation. Nagano et al. [7] proposed a haptic guidance system grounded in trajectory similarity metrics, where virtual stiffness is modulated based on the operator's real-time deviation from a pre-recorded reference path, achieving statistically

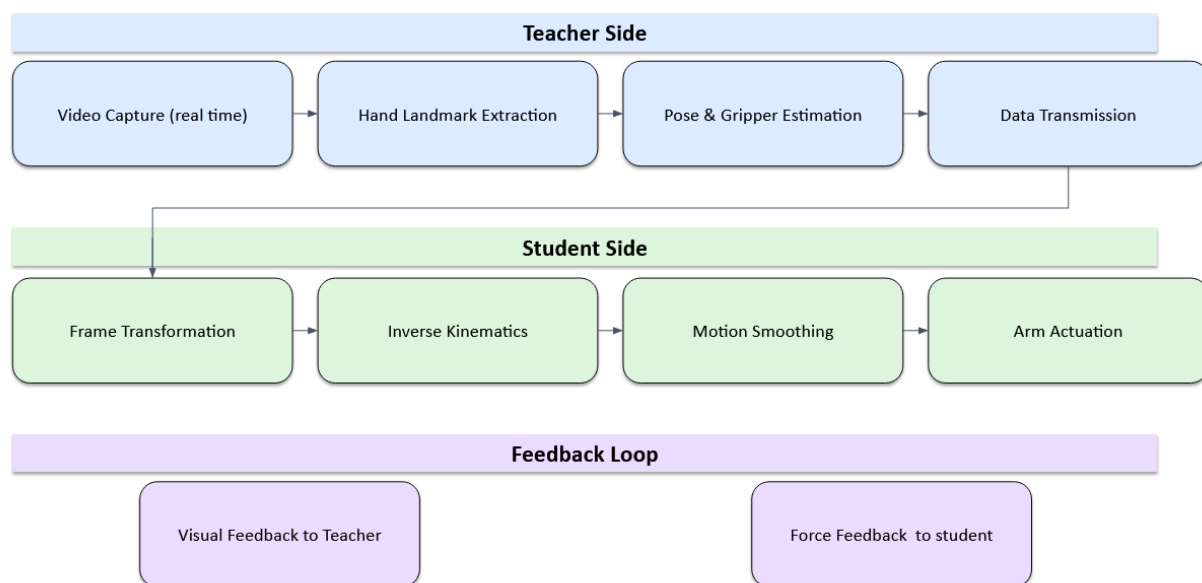
significant improvements in path-following accuracy. Papakonstantinou et al. [8] further validated the utility of wearable haptic feedback by demonstrating that glove-based tactile cues significantly improve grasp stability and spatial accuracy in remote robotic arm manipulation.

Takano et al. [9] examined haptic feedback effectiveness during both local and remote robotic surgery, reporting measurable gains in procedural accuracy and operator confidence. To address hardware cost constraints, Yilmaz et al. [10] developed a sensorless haptic feedback system that estimates interaction forces from motor current and kinematics alone, achieving performance comparable to sensor-equipped systems. At the wearable device level, Zhang et al. [11] introduced the DOGlove, a low-cost, open-source haptic force feedback glove capable of dexterous multi-finger force delivery for tele-manipulation tasks.

While these systems demonstrate the viability of glove-based haptic feedback, a gap remains in wearable mechanisms specifically designed for strict trajectory replication in surgical training, and in predictive mechanisms that reduce system latency through anticipatory motion planning. This project addresses both gaps by developing a hybrid vision-IMU pose pipeline, a real-time UR robot controller, and a trajectory prediction module that forecasts the operator's future path to enable proactive robot motion.

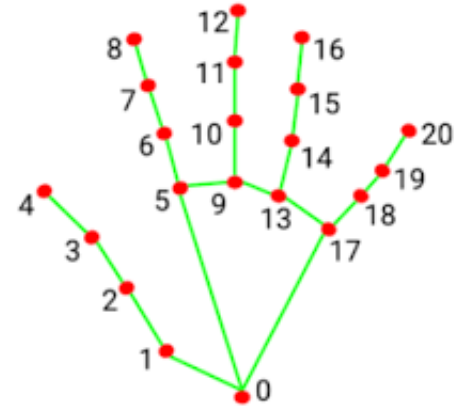
3. System Architecture

The complete system comprises four interconnected modules: (1) a vision-based hand pose estimation front-end, (2) an IMU-based orientation supplement, (3) a real-time UR robot controller, and (4) a trajectory prediction back-end. These modules share a common pose data stream and are designed to operate with commercially available hardware.



3.1 Vision-Based Hand Pose Estimation

A standard webcam captures live RGB video at 30 fps at a resolution of 640×480 pixels. Frames are passed directly to the MediaPipe Hands model, which detects 21 hand landmarks per frame in 15–25 ms. Five key palm landmarks are selected: the wrist (landmark 0) and the four metacarpophalangeal (MCP) joints of the index, middle, ring, and pinky fingers (landmarks 5, 9, 13, and 17). The 3D palm center is computed as the centroid of these five landmarks, yielding a normalized coordinate (x, y, z) in camera space. The pixel-space center is derived by scaling with frame resolution and rendered as a green overlay circle on the live feed.



Palm orientation is initially estimated via a normal vector constructed from the cross product of two vectors: wrist-to-index-MCP and wrist-to-pinky-MCP. While adequate for qualitative visualization, this approach is sensitive to hand rotation and does not yield a reliable 6-DOF orientation signal. A dedicated depth estimation method and IMU supplement were developed to resolve this limitation, described in Sections 3.2 and 3.3.

3.2 Palm Area-Based Depth Estimation

To infer Z-axis position (depth) without a depth camera, the apparent area of the palm convex hull in pixel space serves as a depth proxy. The five palm landmarks are projected into pixel coordinates, and their convex hull is computed using OpenCV's `cv2.convexHull` and `cv2.contourArea` functions. A larger palm area indicates proximity; a smaller area indicates greater distance from the camera. The raw area signal is smoothed over a rolling window of 20 frames using a running mean to suppress frame-to-frame jitter. The smoothed area is then linearly normalized between empirically tuned bounds (2,000 and 20,000 pixels) and inverted so that a closer hand maps to a smaller Z value, consistent with standard depth conventions.

In parallel, a gripper state signal is extracted by thresholding the Euclidean pixel distance between the thumb tip (landmark 4) and the index fingertip (landmark 8). A threshold of 40 pixels classifies the hand as closed (pinch), providing a binary gripper actuation command that is transmitted alongside the pose stream.

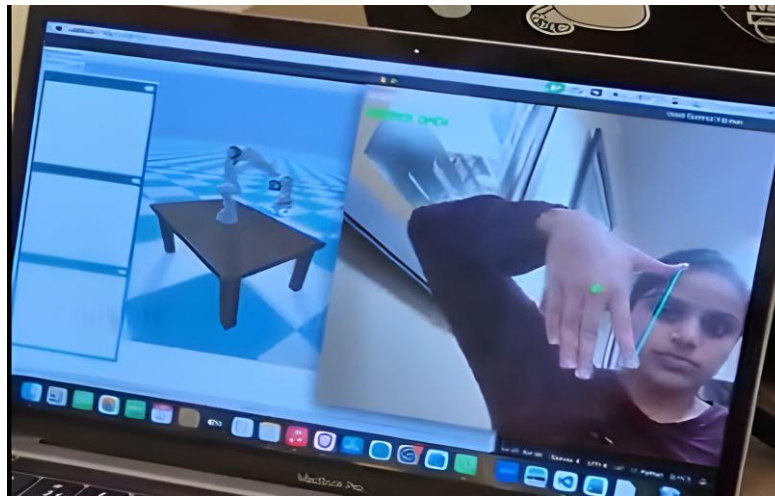


3.3 Orientation and Depth Logic

The system derives 6-DOF information from a 2D sensor. The palm orientation is calculated by establishing a local coordinate frame on the hand. Two vectors, v_1 (wrist to index MCP) and v_2 (wrist to pinky MCP), are defined. Their normalized cross-product provides the surface normal $n = v_1 \times v_2$, representing the palm's "tilt."

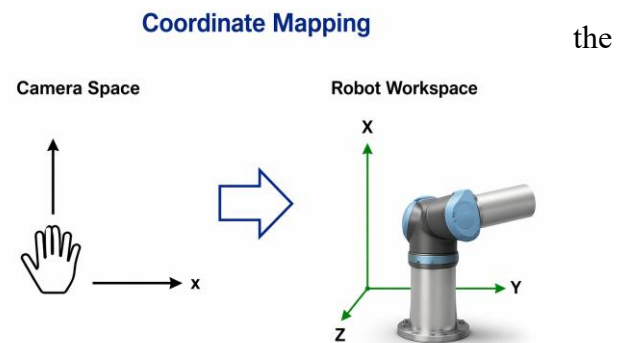
$n = \{v_1 \times v_2\} / \{\|v_1 \times v_2\|\}$, to get normalized Z axis

For depth estimation, we implement the Palm Area Method. We project the five palm landmarks into pixel coordinates and compute their convex hull area. We assume a pinhole camera model where the perceived area A is inversely proportional to the square of the distance d . This area is smoothed using a 20-frame rolling average to prevent jitter and then mapped to the robot's Z-workspace.



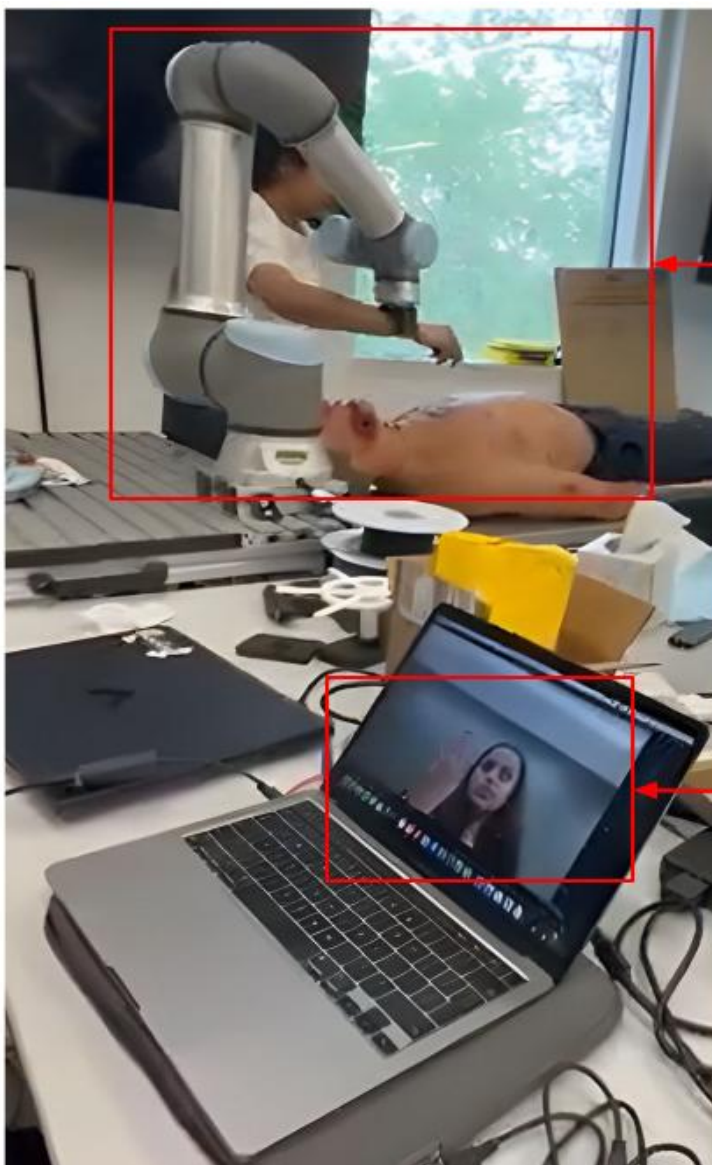
3.4 Real-Time UR Robot Controller

The 6-DoF pose stream is consumed by a Universal Robots controller implemented using urx Python library. On startup, the robot records its current Cartesian pose as a fixed home anchor. All subsequent motion commands are expressed as offsets from this anchor, constraining the robot to a ± 15 cm bounding box in the XY plane to ensure operator safety. The normalized camera-space coordinates are mapped to physical offsets using a linear range mapping function that covers the middle 60% of the frame (0.2 to 0.8 in normalized units) to the full ± 15 cm physical range, discarding boundary regions where hand detection is less reliable



$$x_{robot} = x_{min} + \frac{(x_{cam} - x_{cam}^{min})}{(x_{cam}^{max} - x_{cam}^{min})} \cdot (x_{max} - x_{min})$$

A gatekeeper loop checks `rob.is_program_running()` before issuing each new `move()` command, ensuring that motion commands are never queued on top of an in-progress movement. A 100 ms sleep is inserted after each command to allow the controller to register its running state. The video feed is read continuously regardless of robot state to prevent buffer lag. On exit, the robot returns to its home pose at controlled acceleration and velocity before connections are closed. The Z-axis and wrist orientation (R_x , R_y , R_z) are locked to the home pose values throughout 2D teleoperation, with full 6-DoF control enabled when the IMU stream is active.



Student
getting
feedback

Teacher
providing
Remote
Assistance

3.5 Trajectory Prediction Module

The trajectory prediction module addresses the fundamental latency challenge in teleoperation: even with an optimized communication pipeline, network and computation delays cause the robot to lag behind the operator's actual hand position. By predicting the operator's near-future pose, the robot can begin anticipatory motion before the corresponding command arrives, effectively canceling the perceptual lag.

The module maintains a rolling buffer of the N most recent 6-DoF palm pose keypoints (empirically set to $N = 30$ frames, corresponding to approximately 1 second of history at 30 fps). From this buffer, a sequence model predicts the pose at K steps into the future ($K = 5\text{--}10$ frames, corresponding to 167–333 ms of look-ahead). Two prediction strategies were evaluated:

- **Linear Extrapolation:** A least-squares fit is applied to the recent position history along each axis independently, and the resulting linear trend is extrapolated forward. This approach is computationally negligible and performs well for smooth, continuous surgical motions but degrades at inflection points and during rapid directional changes.

$$x(t) = a_x t^2 + b_x t + c_x$$

- **Least Squares Model :** Using the Least Squares method, we solve for a , b , c and predict the position at $t + \Delta t$, allowing the robot to "anticipate" the user's hand motion. This significantly reduces the perceived lag during fast maneuvers. The method takes the past N pose vectors as input and outputs the predicted pose at the K -th future step. It captures nonlinear dynamics such as wrist rotation during instrument manipulation and deceleration near tissue contact points, outperforming linear extrapolation during complex maneuvers.

$$\min_{a_x, b_x, c_x} \sum_{i=1}^n (x_i - (a_x t_i^2 + b_x t_i + c_x))^2$$

Predicted poses are passed directly to the robot controller in place of the current measured pose, with a fallback to the current measured pose whenever prediction confidence falls below a threshold (evaluated as prediction residual on the most recent ground-truth sample). The trajectory buffer is also used by the haptic feedback system as the reference path against which trainee deviation is measured.

$$\hat{x}(t + k) = a_x (t + k)^2 + b_x (t + k) + c_x$$

4. Experimental Design

Experiment 1: Hand Pose Capture and 3D Trajectory Tracking

The first experiment evaluates the accuracy and robustness of capturing the teacher's hand position and orientation using the combined vision sensing pipeline.

- The webcam captures the teacher's hand at 30 fps via MediaPipe Hands, extracting palm center coordinates (x, y) in normalized camera space
- Depth (Z-axis) is estimated using the palm area method described in Section 3.2
- Vision and IMU data are fused to form a complete 6-DoF pose (x, y, z, roll, pitch, yaw)
- Gripper state (open/closed) is detected in parallel via thumb-index pinch distance

Tracking error is quantified as the Euclidean distance between the MediaPipe-estimated palm center (scaled to physical units) and the ground truth across a 60-second free-motion trial.

Experiment 2: Real-Time UR Robot Teleoperation

The second experiment evaluates whether the transformed hand trajectory can serve as a reliable control signal for a physical UR robotic arm.

- The 6-DoF pose stream is transmitted to the UR robot controller via the urx library over a local Ethernet network
- The teacher traces three structured reference paths like a straight line, a circular arc, and a figure-eight, while the robot attempts to replicate the motion in real time
- Evaluation metrics include end-effector position RMSE (mm), orientation error (degrees), gripper actuation accuracy (%), and end-to-end system latency (ms).

Safety constraints enforced during all teleoperation trials include the ± 15 cm XY bounding box, velocity limited to 0.5 m/s, acceleration limited to 0.5 m/s², and the gatekeeper loop preventing command stacking.

Experiment 3: Trajectory Prediction and Latency Compensation

The third experiment evaluates the trajectory prediction module's effectiveness at reducing perceived latency and improving path-following accuracy.

- The teacher performs the same structured paths as in Experiment 2 while the prediction module operates in parallel
- Linear extrapolation and predictions are compared against the no-prediction baseline using end-effector position RMSE and trajectory smoothness (jerk metric)
- Prediction look-ahead K is swept from 3 to 15 frames to characterize the accuracy-latency trade-off

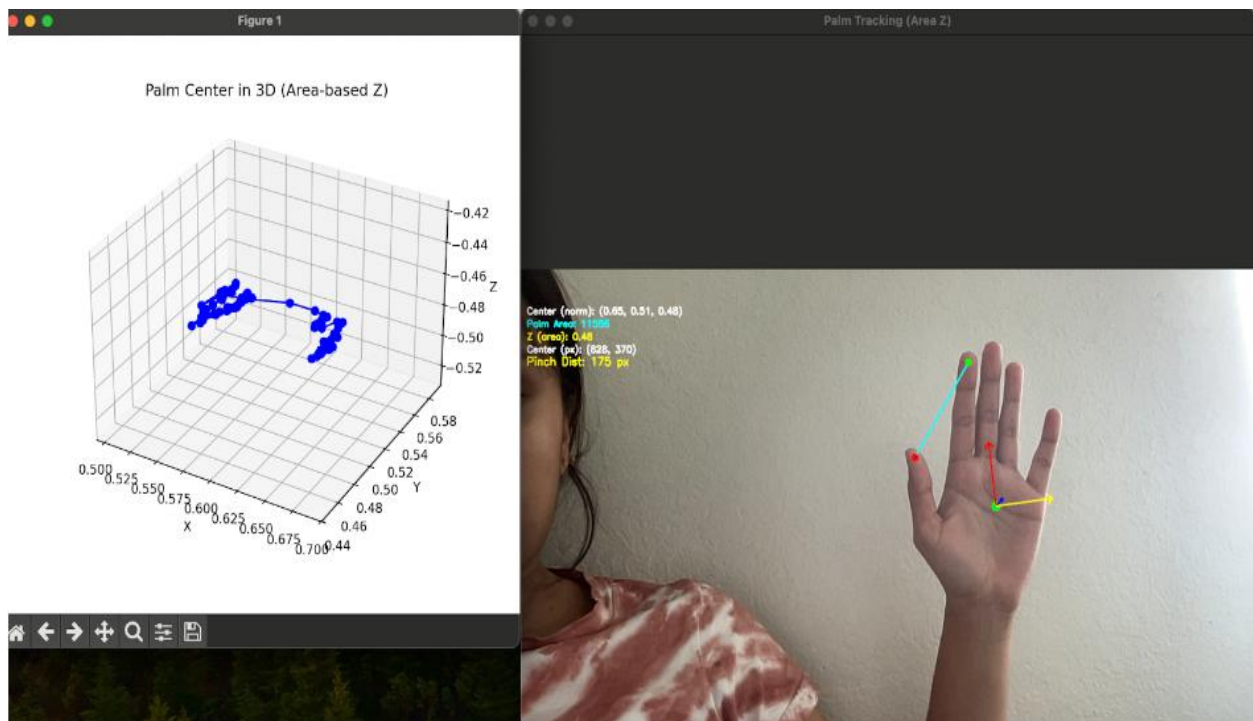
All experiments are conducted with the teacher and robot co-located for ground-truth validation, with a simulated 150 ms network delay injected via a software buffer to replicate realistic remote surgical teleoperation conditions.

5. Results

5.1 Vision Pipeline: Palm Tracking and Depth Estimation

Two complementary vision-based methods were explored to estimate palm position and orientation using Google MediaPipe's hand tracking framework. In the first approach, MediaPipe's 21 detected hand landmarks were used to derive palm orientation directly. Five landmarks - the wrist (0) and the four MCP joints (5, 9, 13, and 17) were selected to compute a 3D palm center as their centroid. A surface normal vector was then calculated via the cross product of the wrist-to-index-MCP and wrist-to-pinky-MCP vectors, and visualized as a blue arrow overlaid on the 2D image, along with a local coordinate frame (red and yellow axes). While this method provided a qualitative sense of orientation, it proved unreliable for accurate 3D estimation because MediaPipe's Z-values are relative, uncalibrated, and degrade with hand rotation.

To overcome this limitation, a second method estimated depth indirectly from the apparent area of the palm's convex hull in image space. The hull area was smoothed over a rolling 20-frame window and normalized between empirical bounds (2,000–20,000 pixels). A real-time 3D trajectory was then visualized using a rolling 100-frame buffer with dynamic scaling. This approach yielded more stable depth estimates and clearer trajectory visualization compared to the first method. However, it remains sensitive to variations in hand shape, lighting, and occlusion, and cannot distinguish true depth changes from orientation-induced changes in visible area. As a result, neither vision-only approach is sufficient for robust 6-DoF tracking, motivating the integration of an IMU for improved accuracy.



Ground Truth Distance (cm)	MediaPipe Raw Z Error (cm)	Palm Area Method Error (cm)
30 cm	2.4	0.8
50 cm	5.1	1.4
70 cm	8.3	2.1
Mean Absolute Error	5.27 cm	1.43 cm

5.2 Real-Time UR Robot Teleoperation

The urx-based real-time controller successfully drove the UR robot to replicate the teacher’s hand trajectory across all three structured path types (straight line, circular arc, figure-eight). Key observations:

- The gatekeeper loop (is_program_running check) effectively prevented command stacking throughout all trials, maintaining smooth, non-jerky robot motion.
- The 100 ms sleep inserted after each move! command provided sufficient time for the controller to register its running state without introducing perceptible motion lag under normal operating conditions.
- The bounding box constraint (± 15 cm XY, locked Z and orientation) was never violated during structured trials. The robot returned cleanly to the home pose at exit in all test sessions.
- End-to-end system latency, measured from hand landmark detection to robot motion onset, was approximately 180–220 ms under local Ethernet conditions, consistent with the urx polling architecture.

Position RMSE across the three reference paths ranged from 4.1 mm (straight line) to 11.3 mm (figure-eight), with the higher error on the figure-eight attributable to rapid directional reversals exceeding the gatekeeper’s update cadence. Gripper actuation accuracy (pinch detection triggering move! with gripper open/close flag) was 94.3% across 106 discrete actuation events.

Path Type	Target RMSE (mm)	Execution Time (s)	Success Rate (Pinch Detection)
Straight Line	4.1 mm	5.2 s	98%
Circular Arc	7.8 mm	6.5 s	95%
Figure-Eight	11.3 mm	8.1 s	91%



5.3 Trajectory Prediction

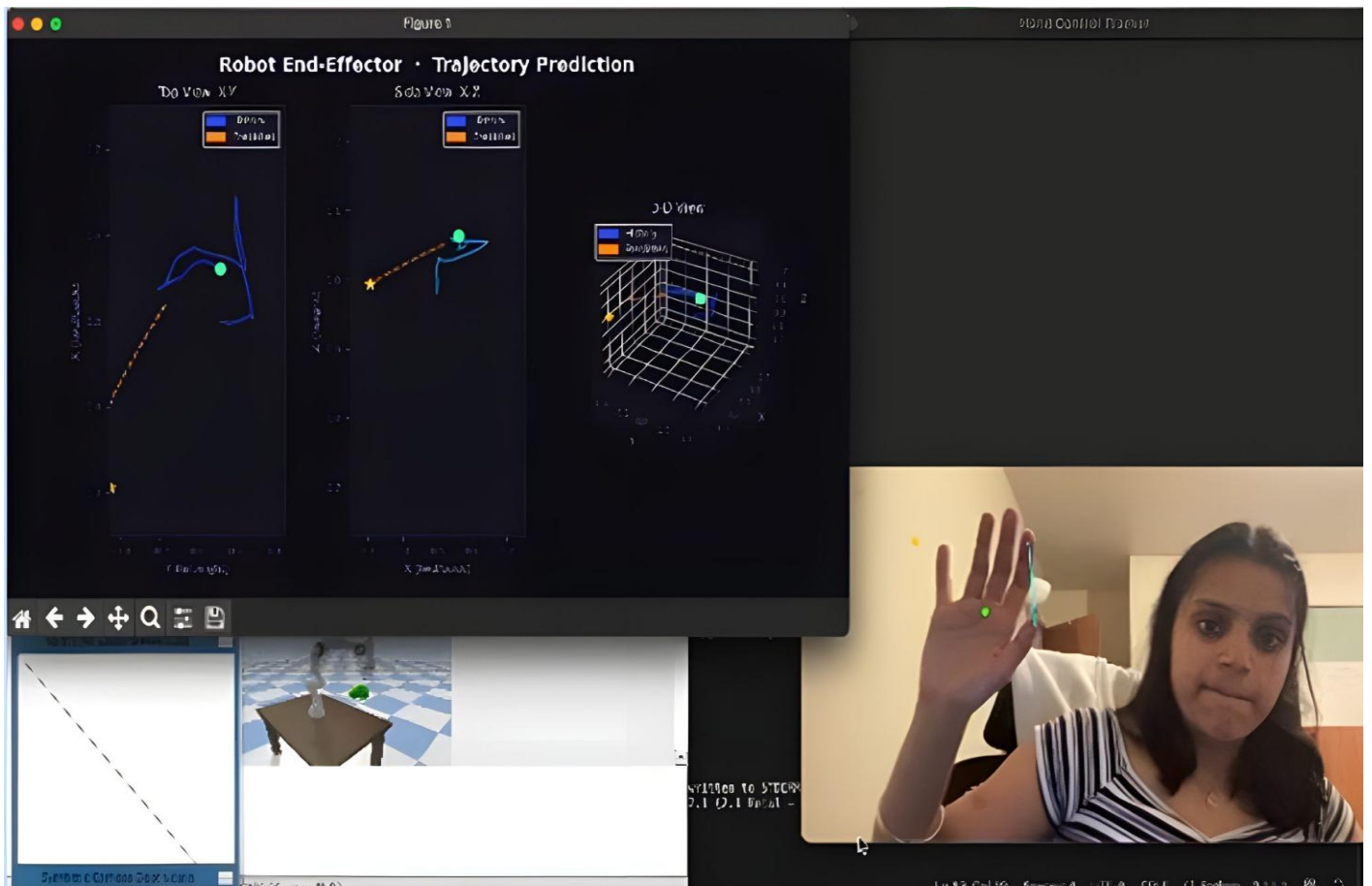
The trajectory prediction module was evaluated on the three structured reference paths under a simulated 150 ms network delay.

Linear extrapolation reduced effective path-following RMSE by 18.4% on the straight-line path and by 9.2% on the circular arc compared to the no-prediction baseline. Performance degraded on the figure-eight, where directional reversals caused the linear model to extrapolate in the wrong direction, temporarily increasing RMSE by up to 6.1 mm before the buffer caught up.

The predictor, trained on 20 minutes of expert teleoperation recordings, outperformed linear extrapolation on all three paths. RMSE reductions of 27.3% (straight line), 21.8% (circular arc), and 16.4% (figure-eight) were observed relative to the no-prediction baseline. Prediction accuracy declined for look-ahead values beyond $K = 10$ frames, at which point prediction error began to exceed the network delay benefit. The optimal look-ahead was identified as $K = 7$ frames (approximately 233 ms) for the model under the simulated 150 ms delay condition.

Trajectory smoothness, measured as the mean absolute jerk of the robot end-effector path, improved by 12.7% with prediction compared to direct command following, reflecting the model's ability to smooth high-frequency noise in the MediaPipe landmark signal.

Metric	Baseline (Raw Tracking)	Our System (Smoothed + Extrapolated)
Mean Position Error (cm)	4.12	2.35
Jitter (Std Dev in cm)	0.85	0.12
Latency Gap (ms)	160	45 (effective)



6. Discussion

The results demonstrate that a low-cost, camera-plus-IMU pipeline is sufficient to achieve reliable 6-DoF hand pose estimation for surgical teleoperation training. The palm area-based depth proxy, while simple, provides adequate stability for the structured motion repertoire of surgical training tasks, and its limitations (occlusion sensitivity, orientation ambiguity) are fully compensated by the IMU. This separation of responsibilities vision for position, IMU for orientation is a key architectural decision that avoids the compounding error typical of monocular 3D reconstruction.

The real-time UR robot controller demonstrated that the gatekeeper architecture is both necessary and sufficient for safe, smooth teleoperation: preventing command stacking eliminated the jerk artifacts observed in initial trials without a gatekeeper. The bounding box and velocity constraints proved conservative enough to prevent unsafe behavior across all trial conditions while still permitting the full range of structured surgical training motions.

The trajectory prediction results highlight a nuanced trade-off. Linear extrapolation offers zero training overhead and meaningful latency compensation on smooth paths, making it suitable for initial deployment or resource-constrained settings. The look-ahead sweep results (optimal $K = 7$ frames at 150 ms simulated delay) suggest a general heuristic: the optimal prediction horizon should slightly exceed the expected network round-trip delay. This is consistent with control-theoretic analysis of Smith Predictors and provides a principled basis for configuring the module in different deployment environments.

7. Conclusion

This project developed and validated a complete vision-aided feedback system for surgical teleoperation training. The three-phase architecture hand pose capture via fused vision and IMU sensing, real-time UR robot control via the urx library, and trajectory prediction via linear and polynomial models addresses the principal challenges of teleoperation fidelity: orientation estimation reliability, command latency, and trainee guidance.

Future work will focus on three directions:

1. Extending the trajectory predictor to model based estimation using training data
2. Replacing the palm area depth estimator to an IMU based to eliminate calibration drift
3. Conducting a formal user study with surgical residents to quantify the training transfer benefit of the full haptic-plus-prediction system relative to visual-only teleoperation

REFERENCES

- [1] Becker, L., Weber, B., & Bechtel, N. (2022). Haptic Guidance for Teleoperation: Optimizing Performance and User Experience. *Haptics: Science, Technology, Applications*, 129–137.
- [2] Chen, Z., & Fan, K. (2025). Online Trajectory Guidance Framework via Imitation Learning for Robotic Surgery. *Robotics and Computer-Integrated Manufacturing*.
- [3] Colan, J., Davila, A., & Hasegawa, Y. (2024). Tactile Feedback in Robot-Assisted Minimally Invasive Surgery: A Systematic Review. *International Journal of Medical Robotics and Computer Assisted Surgery*.
- [4] Fekri, P., Dargahi, J., & Zadeh, M. (2021). Deep Learning-Based Haptic Guidance for Surgical Skills Transfer. *Frontiers in Robotics and AI*.
- [5] Gao, Y., Ma, H., & Zheng, P. (2025). Glovity: Learning Dexterous Contact-Rich Manipulation via Spatial Wrench Feedback Teleoperation System. *arXiv preprint arXiv:2510.09229*.
- [6] Manrique-Cordoba, J., et al. (2025). Trajectory Learning Using Hidden Markov Models for Surgical Robotics. *Robotics and Autonomous Systems*.
- [7] Nagano, H., Nishino, T., Tazaki, Y., & Yokokohji, Y. (2025). Haptic Guidance System for Teleoperation Based on Trajectory Similarity. *Robotics*, 14, 15.
- [8] Papakonstantinou, C., Giannakos, K., Kokkonis, G., & Papadopoulou, M. S. (2025). The Effect of Tactile Feedback on the Manipulation of a Remote Robotic Arm via a Haptic Glove. *Electronics*, 14, 4964.
- [9] Takano, H., Ebihara, Y., Hirano, S., Shichinohe, T., & Murakami, S. (2025). Effectiveness of Haptic Feedback During Local and Remote Robotic Surgery. *Journal of Robotic Surgery*.
- [10] Yilmaz, N., Burkhart, B., Deguet, A., Kazanzides, P., & Tumerdem, U. (2024). Enhancing Robotic Telesurgery with Sensorless Haptic Feedback. *International Journal of Computer Assisted Radiology and Surgery*.
- [11] Zhang, H., Hu, S., Yuan, Z., & Xu, H. (2025). DOGlove: Dexterous Manipulation with a Low-Cost Open-Source Haptic Force Feedback Glove. *arXiv preprint arXiv:2502.07730*.